HATI: Current State and Forward Plan

A status and roadmap report on a physics-based, two-channel lunar terrain-hazard pipeline, prepared for external review

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Máni Mission terrain-hazard pipeline (HATI)

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Abstract

This report is a candid account of where HATI stands and where it is going, written to be picked apart. HATI is a lunar terrain-hazard pipeline built from two interpretable, training-free channels: a multi-scale roughness *heatmap* on a stereo elevation model, and a *shadow census* that recovers obstacles below the elevation model's resolution from a finer photograph. Its design bet is that a landing system needs a detector it can *audit*, not the most accurate black box. We report what is established on real data: the heatmap is a within-scene relative hazard ranker (not a calibrated probability); a resolution ladder gives the channels their first measured detection numbers (shadow AUC 0.83–0.99; heatmap 0.86–0.93); a controlled cross-site test showed, against our earlier hope, that eight of the ten roughness *texture* channels score *below chance* between sites and have been removed, leaving the two *physical* channels that do transfer, RMS slope and terrain ruggedness (cross-site AUC 0.826 and 0.808); and a deterministic slope gate built on those physical channels is conservative and specific ($\sim$5 % false alarm on benign mare at an 8° limit) yet correctly passes the gentle (3°) Athena touchdown, confirming that the Athena-class hazard is sub-resolution and belongs to the shadow channel. We then report a new sub-resolution channel, *shadow kinematics*, which uses the Sun's motion over a solar sweep to separate real obstacles from albedo and to measure their height; it is proven on synthetic ground truth (height RMS error 0.24 m; albedo decoys driven to zero confidence) but *not yet on real imagery*. The forward plan, with rationale and method, is: ingest an azimuth-diverse sweep (already shown to exist, with 525 usable frames over the site) to produce the first real-data kinematics result; fuse the gross-terrain gate and the sub-resolution obstacle layer into one calibrated hazard map; and validate on a held-out site with frozen thresholds. We mark clearly what is proven on real data, what is proven only in simulation, and what is planned, and we end with the specific questions we would most like a reviewer to press on.

Keywords: lunar landing hazards; sub-resolution roughness; shadow photoclinometry; multi-illumination; interpretable detection; Mons Mouton; terrain-relative navigation.

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1 Motivation and design stance

On 6 March 2025 the IM-2 spacecraft set down near 84.79° S, 29.20° E on Mons Mouton and tipped over. The post-landing analysis tied the upset to a crater roughly 20 m across on sloped ground at the touchdown, a feature at or below the floor of the elevation models available before the flight. HATI exists to answer a narrow, testable question: using only pre-landing orbital data, can an interpretable pipeline flag the class of hazard that this represents?

HATI carries two channels by design, because the hazard lives at two scales and no single map sees both. Gross terrain, the slopes and relief a lander cannot straddle, is visible in a digital elevation model (DEM). Sub-metre-to-metre obstacles, the rocks, shallow craters and the Athena crater itself, are not; they must be read from a finer photograph, through the shadows they cast. The binding design constraint is auditability: every flag must trace to a stated physical quantity a reviewer (or a flight-safety board) can replay. That rules out a black-box classifier as the decision layer and is the reason the pipeline is deterministic and training-free.

In plain words

Two problems, two tools. An elevation map catches the big stuff: steep walls, large craters. It is blind to a boulder or a shallow pit smaller than its pixels, which is exactly what tipped Athena. For those you read the *shadows* in a sharper photo. HATI runs both and insists that every warning it raises can be explained in plain physical units, because a lander's safety case cannot rest on "the network said so."

2 Current architecture

Channel 1: physical gross-terrain gate. On a stereo DEM we compute roughness descriptors at explicit physical baselines, robustly normalise them (log transform, median/IQR z-score, clipped), and fuse them with *fixed* literature-anchored weights, never fitted, so the score decomposes into exact signed per-channel contributions (Kreslavsky and Head 2000; Rosenburg et al. 2011; Kreslavsky et al. 2013; Cai and Fa 2020). Ten descriptors were evaluated; the cross-site control of section 3.3 retained two, RMS slope and the terrain ruggedness index (Riley et al. 1999), and removed the rest. A deterministic gate then flags ground whose footprint-scale slope exceeds a stated lander limit, in degrees, so the output is comparable between sites rather than relative to one scene.

Channel 2: shadow census. On a co-registered orthophoto (0.9 m/px) (Robinson et al. 2010; Henriksen et al. 2017) we detect shadows by a local-relative brightness threshold, fit each shadow region, and convert its along-Sun length L to an obstacle height by $h = L \tan e$ at the known solar elevation e. At the polar site the Sun grazes the horizon, so a small obstacle throws a long, countable shadow.

There is no third channel. An earlier revision carried the full ten-descriptor roughness stack as a separate texture layer; section 3.3 removed it. Each remaining channel covers one scale regime, and each has cross-site evidence behind it.

3 What is established: on real data

3.1 The Athena forensic test

Running the original ten-descriptor stack on pre-landing (2012) NAC data of the touchdown, the heatmap ranks the actual landing point in the roughest few percent of an already-rough massif (robust $z = 3.22$), but on *texture*, not slope: the local tilt is a gentle 2.8°. Channel 2 finds of order 1400 sub-resolution obstacles per km^2 that the DEM cannot see. Two readings matter here, and they pull in different directions. The z figure establishes the stack as a *within-scene relative ranker*, never a calibrated probability of loss of mission. It also predates the cross-site control, and eight of the descriptors producing it have since been removed (section 3.3), so the $z = 3.22$ result should be read as a statement about that one scene and not as evidence that texture is diagnostic. The absolute reading, that this ground measures 3.0° and is not steep, is the one that transfers.

3.2 First measured detection numbers

A resolution ladder (degrade real data, run each channel on the coarse version, score against the hazards visible at full resolution) gives HATI its first ROC numbers. The shadow channel holds AUC $0.99/0.97/0.94/0.88/0.83$ at $1.8/2.7/3.6/5.4/7.2$ m with a quantified size-reach; the heatmap recovers resolved relief on an independent 1.5 m DEM of Apollo 17 at AUC $0.86/0.92/0.93$ as posting coarsens to $3/6/12$ m.

3.3 The cross-site control, and the channels it removed

We tested whether the heatmap is *absolute* by calibrating it on genuinely smooth terrain (the Apollo 11 mare) and asking whether it separates that mare from the rough massif. It did not, in the way we first hoped, and chasing that down changed the project.

A third round of the control settled the question by separating two candidate explanations that earlier rounds had conflated: an *effective-resolution* mismatch between the two scenes, and channel definitions expressed in pixels rather than metres. Three configurations were scored on identical pixel sets (205 660 massif, 299 273 mare): (A) posting matched only, which reproduces the earlier result; (B) posting and effective resolution both matched; (C) as B, with every channel redefined at explicit physical baselines. Table 1 gives the outcome.

Table 1: Cross-site AUC per channel, massif ranked above mare. Below 0.50 a channel ranks the *smooth* site above the *rough* one, which is worse than uninformative.

Channel	A posting	B eff. res.	C physical	Disposition
RMS slope	0.789	0.817	**0.826**	retained
TRI	0.776	0.808	**0.808**	retained
Slope IQR	0.434	0.438	0.432	removed
|TPI|	0.420	0.420	0.420	removed
MDS 16 m to 64 m	0.317–0.376	0.351–0.375	0.351–0.375	removed
Curvature IQR	0.215	0.369	0.369	removed
Planar deviation	0.305	0.342	0.342	removed

Two channels cleared a 0.70 bar and eight did not. The eight that failed sit below 0.50, so they rank smooth ground above rough ground more often than not. Matching effective resolution (A to B) is the only correction that moved anything materially, and it moved only curvature IQR, by $+0.153$; that channel nevertheless finished at 0.369, having become markedly less wrong without becoming useful. Redefining channels physically (B to C) changed the non-MDS channels by essentially nothing, which is expected here because the common posting had already made the pixel windows equivalent. Mean texture-channel AUC across all fixes moved from 0.349 to 0.377.

The two survivors have a property in common: RMS slope is measured in degrees and TRI in metres. On those physical channels the Athena touchdown is *benign* in absolute terms (slope 2.74° against a mare median near 2.3°; relief 3.08 m below the mare's 8.7 m 95th percentile).

Why it is built this way

This is where the project changed direction, so we state it plainly rather than bury it. Our first instinct, that a per-scene texture score is a portable, absolute hazard map, was wrong, and a controlled experiment showed it. The correction is not a patch. The eight failing channels have been *removed* from the operational model rather than down-weighted, because a channel scoring below chance on a controlled cross-site test cannot be justified in a safety system at any weight. HATI is therefore a two-channel pipeline: a physical gross-terrain gate built on RMS slope and TRI, and the sub-resolution shadow channel. We would rather show the reviewer the negative result, and act on it, than present a polished overclaim.

3.4 The absolute gate, and where Athena lands under it

Built on the transferable physical channels, the deterministic slope gate is both conservative and specific: at an 8° footprint-slope limit it raises a ∼5.3 % false alarm on benign mare (an upper bound, since the mare has real crater walls) versus the ∼20 % the texture model produced, and flags 14 % of the Mons Mouton scene. The Athena touchdown, at 3.0° footprint slope, *passes* the gate, correctly. The gate is doing its job: it rules on gross terrain and reports, truthfully, that the

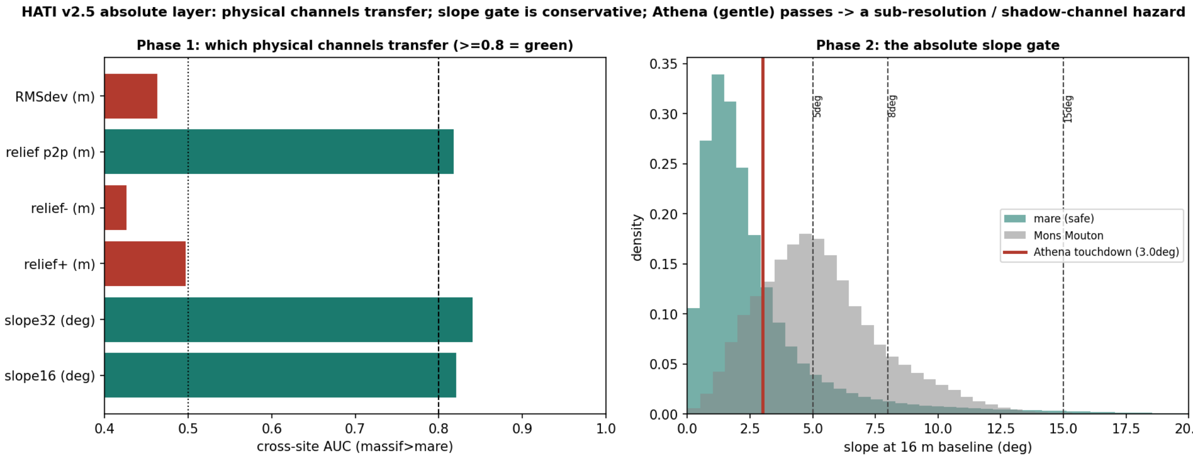

Figure 1: The absolute layer, on real data. **Left:** which physical channels transfer between a smooth mare and the rough massif (green ≥ 0.80); slope and relief-amplitude transfer, texture channels do not. **Right:** the deterministic slope gate, showing the mare and massif slope distributions on one physical scale, the nominal 8° lander limit, and the Athena touchdown at 3.0°, which passes.

touchdown is not grossly hazardous. What tipped Athena is below its resolution. This is the division of labour the whole pipeline now rests on (figs. 1 and 2).

4 What is established: in simulation only

4.1 Shadow kinematics: the sub-resolution channel

A single shadow photograph cannot tell a real shadow from a dark rock, cannot fix an obstacle's sign or height without assuming flat ground, and only sees obstacles lit to cast toward the camera under one Sun. The shadow-kinematics channel removes all three by treating the Sun as a moving probe over a *solar sweep* (many orthophotos of one site under different solar azimuths). A true obstacle's shadow pivots about a fixed base while its length tracks the elevation; a dark albedo mark does not move. Accumulating one vote at the up-Sun base of every detected shadow makes a real obstacle's votes *converge* (its height falls out of the length statistics) while an albedo mark's votes *smear* and self-cancel (Bruce Hapke 1984; Sato et al. 2014).

On a synthetic scene with known truth (70 obstacles of 0.3 m to 3 m with sub-resolution footprints, plus 40 adversarial albedo and streak decoys, rendered under a 0° to 360° sweep at 5° elevation), the detector recovers heights to an RMS error of 0.24 m, holds detection AUC at 0.99, and drives decoy confidence to zero while genuine obstacle confidence accumulates (a separation of order 8000 at full sweep; fig. 3). We are careful about the claim: the win is on *height, false-alarm rate and completeness*, not on the detection AUC, which saturates because the orientation gate alone rejects round decoys in a single frame.

4.2 The sweep exists

The method needs azimuth diversity, so we checked the archive is adequate before committing to ingestion. A query of the NASA Orbital Data Explorer over the Athena site returns 696 LROC narrow-angle frames, 525 with the Sun above the horizon, spanning 2010 to 2024 and filling all twelve 30° azimuth sectors. The sweep is real and over-determined; what is missing is the co-registration to turn those frames into a usable stack.

5 Evidence ledger

The single most useful thing for a reviewer is an honest map of claim to evidence (table 2). "Real" means measured on flight or archival data; "Synthetic" means demonstrated in simulation with

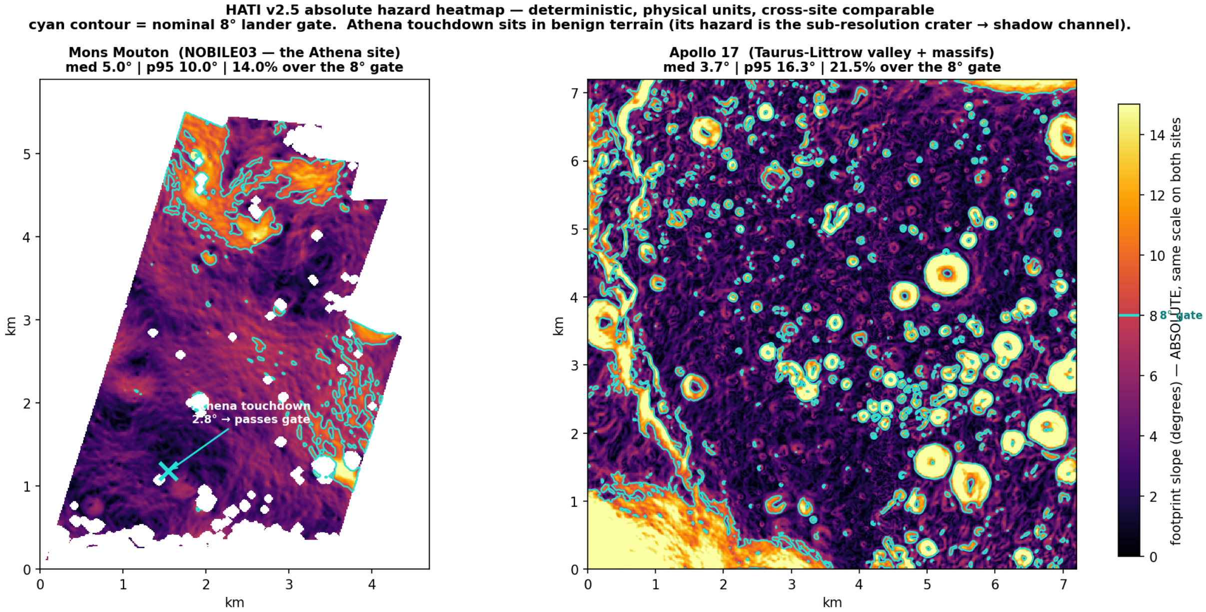

Figure 2: Because the channel is in physical units at matched resolution, two sites are directly comparable on one colour scale: a 7° slope is 7° on both. Mons Mouton (the Athena site) with the touchdown marked, beside Apollo 17's valley and massifs; the cyan contour is the 8° gate. A per-scene z-score heatmap cannot support this comparison.

known truth; "Planned" means designed but not yet run.

6 Forward plan: why and how

The plan has one milestone that matters and three that package it. We give each a rationale and a method.

6.1 Real-data shadow kinematics (the milestone)

Why. Until the kinematics detector runs on a real azimuth-diverse sweep, HATI's central claim, that it senses the sub-resolution hazards the maps miss, is proven only in simulation. This is the step that turns a promising estimator into a result on the Moon. *How.* Build an ingestion pipeline that selects an azimuth-spread subset of the 525 lit frames, calibrates and map-projects each (ISIS: `lronac2isis` → web `spiceinit` → `lronaccal` → `cam2map` to polar stereographic), and sub-pixel co-registers them to the reference grid. The co-registration error budget is the real risk: a mislocated shadow base blurs the confidence peak and biases the height, so quantifying that budget on real frames is the first task, not an afterthought.

6.2 The integrated absolute hazard map

Why. The gross-terrain gate and the sub-resolution obstacle layer are still separate scripts; a lander needs one map. *How.* Combine the deterministic slope/relief gate with the kinematics obstacle density through a threshold-anchored noisy-OR, then calibrate the result against outcomes so the output is a defensible strike probability of Poisson form, $P = 1 - \exp(-\lambda A)$, with λ the density of obstacles above the lander's clearance and A the footprint area, reported honestly as a relative hazard index until enough ground truth exists to calibrate it into an absolute one.

6.3 Held-out validation

Why. Every number above is either within-scene or on sites used to reason about the method. The credible test is generalisation. *How.* Freeze the gates and thresholds, then run the whole

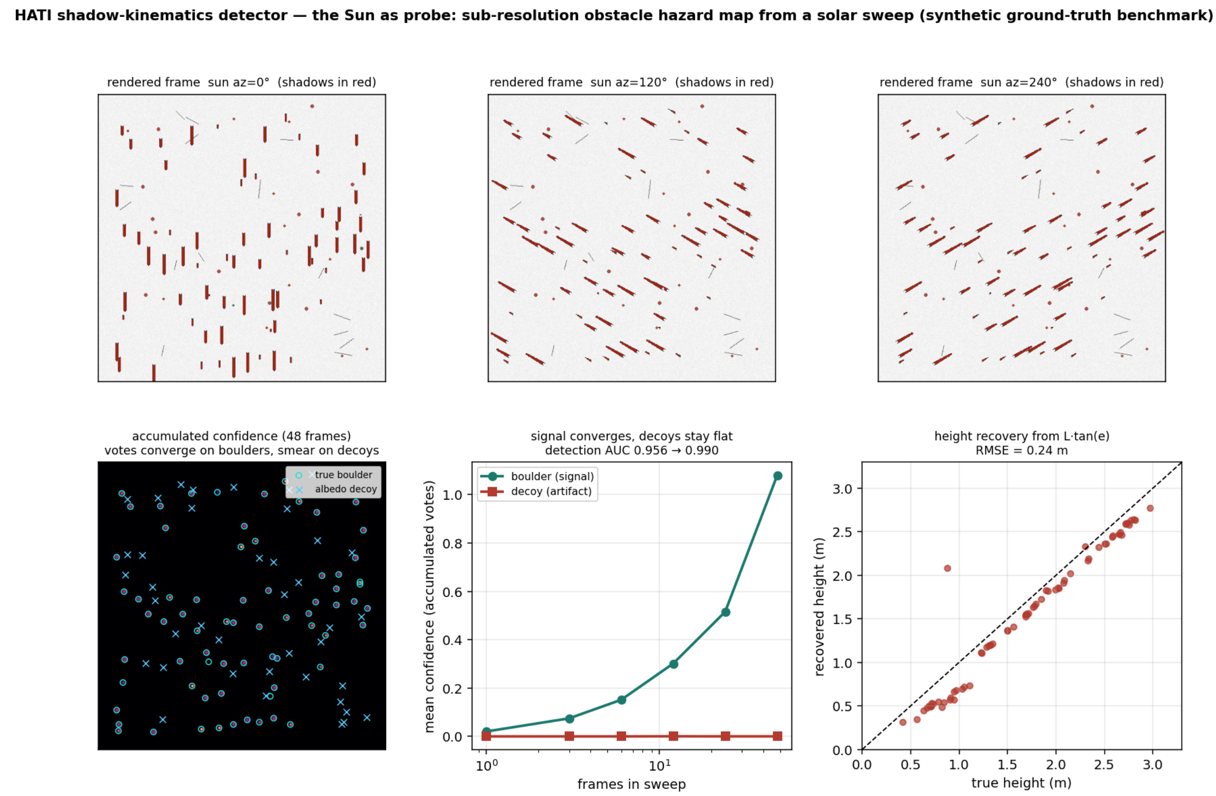

Figure 3: Shadow kinematics on synthetic ground truth. **Top:** one scene under solar azimuths 0°, 120°, 240°, where the shadows (red) pivot about their casters. **Lower left:** accumulated confidence after 48 frames converges on the true obstacles (cyan) and never settles on the decoys (blue). **Lower middle:** obstacle confidence grows with sweep size while decoy confidence stays at zero. **Lower right:** recovered vs true height, RMS error 0.24 m.

pipeline untouched on a site it has never seen, and report the detection and false-alarm rates that result. No knob may move after freezing.

6.4 Capstone and portable core

Why. The work is currently spread across a v2.0 architecture paper, a shadow-kinematics note, and this report; it needs one honest end-to-end account, and the detection engine needs to be a clean module rather than a pile of scripts. *How.* Fold the cross-site reorientation, the absolute gate, the kinematics channel with its real result, and the fused map into a single v2.5 paper, and factor the detector into a reusable core.

6.5 Hardening (not blocking)

A Diviner rock-abundance cross-check of the shadow-derived obstacle density, and an independent higher-resolution validation, would harden the case but are not required to call the method established.

Honest limit

The load-bearing gaps, stated for the reviewer. (1) The sub-resolution channel is unproven on real imagery; everything about it rests on a synthetic benchmark that is favourable by construction (flat background, two decoy types). (2) There is no calibrated probability anywhere in the pipeline yet: the heatmap is a relative ranker and the strike probability is not yet tied to outcomes. (3) All cross-site numbers involve sites used to develop the method;

Table 2: What is proven, and how.

Claim / capability	Evidence	Metric
Heatmap ranks the Athena touchdown high *within its scene*	Real	$z = 3.22$, worst few %
Heatmap is a relative ranker, *not* a calibrated P(loss)	Real (by control)	n/a
Shadow census finds sub-resolution obstacles at the site	Real	$\sim$1400/km^2
Resolution-ladder recovery (shadow / heatmap)	Real	AUC 0.83–0.99 / 0.86–0.93
Eight texture channels score below chance and were removed	Real (negative)	AUC 0.342–0.432
Physical channels *do* transfer between sites	Real	AUC 0.82–0.84
Absolute slope gate: conservative + specific	Real	mare FA 5.3 % @ 8°
Athena passes the gross-terrain gate (is sub-resolution)	Real	3.0° slope
Shadow kinematics separates obstacles from albedo	Synthetic	decoy conf. $\to 0$
Shadow kinematics measures obstacle height	Synthetic	RMSE 0.24 m
An azimuth-diverse sweep exists over the site	Real (archive)	525 lit frames, 12/12 sectors
Real-data shadow-kinematics obstacle map	**Planned**	n/a
Fused, calibrated absolute hazard map	**Planned**	n/a
Site-held-out validation with frozen gates	**Planned**	n/a

the held-out test is not yet done. (4) The shadow height law assumes locally flat ground; real slopes bias it, and correcting that needs the same multi-illumination stack we have not yet ingested.

7 Questions we would most like pressed

We are sending this out precisely to have it challenged. The critiques we would value most:

- Is the up-Sun-base voting estimator biased in a way the synthetic test hides, for example on real terrain where shadows fall across slopes rather than flat ground?
- Is the noisy-OR the right fusion for combining a deterministic gate with a probabilistic obstacle layer, or does it double-count correlated evidence?
- Is a resolution ladder on degraded orbital data a fair proxy for real detection performance, or does anti-aliased degradation flatter the method?
- What ground truth would make a *calibrated* strike probability defensible, given how few instrumented lunar landing outcomes exist?
- Is the cross-site transfer of slope and relief (AUC ~ 0.8) strong enough to anchor an absolute gate, or is 0.8 simply not good enough for a safety-critical call?

Reproducibility. The channels, controls, gate, sweep query and kinematics detector are the scripts `athena_counterfactual.py`, `mare_control_v2.py`, `hati25_absolute_gate.py`, `solar_sweep_query.py` and `shadow_kinematics.py`; each is deterministic given its stated inputs. Companion documents: the v2.0 architecture paper and the shadow-kinematics method note.

References

Cai, Yu and Wenzhe Fa (2020). "Meter-scale topographic roughness of the Moon: The effect of small impact craters". In: *Journal of Geophysical Research: Planets* 125.8, e2020JE006429. DOI: 10.1029/2020JE006429.

Hapke, Bruce (1984). "Bidirectional reflectance spectroscopy: 3. Correction for macroscopic roughness". In: *Icarus* 59.1, pp. 41–59.

Henriksen, M. R., M. R. Manheim, K. N. Burns, et al. (2017). "Extracting accurate and precise topography from LROC narrow angle camera stereo observations". In: *Icarus* 283, pp. 122–137.
Kreslavsky, M. A. and J. W. Head (2000). "Kilometer-scale roughness of Mars: Results from MOLA data analysis". In: *Journal of Geophysical Research: Planets* 105.E11, pp. 26695–26711.
Kreslavsky, M. A., J. W. Head, G. A. Neumann, et al. (2013). "Lunar topographic roughness maps from Lunar Orbiter Laser Altimeter (LOLA) data: Scale dependence and correlation with geologic features and units". In: *Icarus* 226.1, pp. 52–66.
Riley, Shawn J., Stephen D. DeGloria, and Robert Elliott (1999). "A terrain ruggedness index that quantifies topographic heterogeneity". In: *Intermountain Journal of Sciences* 5.1–4, pp. 23–27.
Robinson, M. S. et al. (2010). "Lunar Reconnaissance Orbiter Camera (LROC) instrument overview". In: *Space Science Reviews* 150.1–4, pp. 81–124. DOI: 10.1007/s11214-010-9634-2.
Rosenburg, M. A., O. Aharonson, J. W. Head, et al. (2011). "Global surface slopes and roughness of the Moon from the Lunar Orbiter Laser Altimeter". In: *Journal of Geophysical Research: Planets* 116, E02001.
Sato, H., M. S. Robinson, B. Hapke, et al. (2014). "Resolved Hapke parameter maps of the Moon". In: *Journal of Geophysical Research: Planets* 119.8, pp. 1775–1805. DOI: 10.1002/2013JE004580.